

FIFA World Cup Analysis

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Subject Name: Analytics Programming

GitHub Repository:

<https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission>

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Question 1: Data Inspection, Cleaning and Home Penalty Distribution

Rationale

I joined the four datasets because the information for this analysis was not stored in only one file. The `Matches` dataset contains the match results and scores, while the other datasets provide the team, stadium and tournament information. Joining them made it easier for me to use all the information in one data frame.

I changed "?" to NA because "?" is only a text value, but NA is recognised by R as a missing value. I changed `Result`, `Stage`, `Country` and `ExtraTime` to factors because these variables contain categories. I changed `HomePenalty` and `AwayPenalty` to numeric because they show the number of penalty goals.

I used a bar chart for `HomePenalty` because penalty scores are separate whole numbers. The chart makes it easy to see which penalty score appeared more often and which score appeared less often.

Load and Join the Data

```
# Read the data CSV files.
matches <- read_csv("../data/Matches.csv")
stadiums <- read_csv("../data/Stadiums.csv")
teams <- read_csv("../data/Teams.csv")
tournaments <- read_csv("../data/Tournaments.csv")

# Change every "?" value to NA.
matches[matches == "?"] <- NA
stadiums[stadiums == "?"] <- NA
teams[teams == "?"] <- NA
tournaments[tournaments == "?"] <- NA

# Join team information for the home team and away team.
df <- matches %>%
  left_join(
    teams %>%
      rename(
        HomeTeamID = TeamID,
        HomeTeamName = TeamName,
        HomeTeamCode = TeamCode
      ),
    by = "HomeTeamID"
  ) %>%
  left_join(
    teams %>%
      rename(
        AwayTeamID = TeamID,
        AwayTeamName = TeamName,
        AwayTeamCode = TeamCode
      ),
    by = "AwayTeamID"
  ) %>%
```

```
left_join(stadiums, by = "StadiumID") %>%
left_join(tournaments, by = "TournamentID")
```

Inspect and Clean the Data

```
# Display the first six rows.
head(df)
```

```
## # A tibble: 6 x 24
##   MatchID TournamentID MatchName Stage MatchDate MatchTime StadiumID HomeTeamID
##   <chr>      <chr>      <chr>    <chr> <chr>      <time>      <chr>      <chr>
## 1 M-1930-- WC-1930      France v~ grou~ 7/13/1930 15:00      S-193      T-28
## 2 M-1930-- WC-1930      United S~ grou~ 7/13/1930 15:00      S-192      T-80
## 3 M-1930-- WC-1930      Yugoslav~ grou~ 7/14/1930 12:45      S-192      T-84
## 4 M-1930-- WC-1930      Romania ~ grou~ 7/14/1930 14:50      S-193      T-59
## 5 M-1930-- WC-1930      Argentin~ grou~ 7/15/1930 16:00      S-192      T-03
## 6 M-1930-- WC-1930      Chile v ~ grou~ 7/16/1930 14:45      S-192      T-13
## # i 16 more variables: AwayTeamID <chr>, HomeTeamScore <dbl>,
## #   AwayTeamScore <dbl>, ExtraTime <dbl>, PenaltyShootout <dbl>,
## #   HomePenalty <chr>, AwayPenalty <chr>, Result <chr>, HomeTeamName <chr>,
## #   HomeTeamCode <chr>, AwayTeamName <chr>, AwayTeamCode <chr>,
## #   StadiumName <chr>, City <chr>, Country <chr>, TournamentName <chr>
```

```
# Check the number of rows and columns.
dim(df)
```

```
## [1] 964 24
```

```
# Check the variable types.
str(df)
```

```
## spc_tbl_ [964 x 24] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ MatchID      : chr [1:964] "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
## $ TournamentID : chr [1:964] "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
## $ MatchName     : chr [1:964] "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "R
## $ Stage         : chr [1:964] "group stage" "group stage" "group stage" "group stage" ...
## $ MatchDate     : chr [1:964] "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
## $ MatchTime     : 'hms' num [1:964] 15:00:00 15:00:00 12:45:00 14:50:00 ...
## ..- attr(*, "units")= chr "secs"
## $ StadiumID     : chr [1:964] "S-193" "S-192" "S-192" "S-193" ...
## $ HomeTeamID    : chr [1:964] "T-28" "T-80" "T-84" "T-59" ...
## $ AwayTeamID    : chr [1:964] "T-44" "T-06" "T-09" "T-54" ...
## $ HomeTeamScore : num [1:964] 4 3 2 3 1 3 4 3 1 1 ...
## $ AwayTeamScore : num [1:964] 1 0 1 1 0 0 0 0 0 0 ...
## $ ExtraTime     : num [1:964] 0 0 0 0 0 0 0 0 0 0 ...
## $ PenaltyShootout: num [1:964] 0 0 0 0 0 0 0 0 0 0 ...
## $ HomePenalty   : chr [1:964] NA "0" "0" "0" ...
## $ AwayPenalty   : chr [1:964] "0" "0" "0" NA ...
## $ Result        : chr [1:964] "home team win" "home team win" "home team win" "home team win" ...
## $ HomeTeamName  : chr [1:964] "France" "United States" "Yugoslavia" "Romania" ...
```

```
## $ HomeTeamCode : chr [1:964] "FRA" "USA" "YUG" "ROU" ...
## $ AwayTeamName : chr [1:964] "Mexico" "Belgium" "Brazil" "Peru" ...
## $ AwayTeamCode : chr [1:964] "MEX" "BEL" "BRA" "PER" ...
## $ StadiumName : chr [1:964] "Estadio Pocitos" "Estadio Gran Parque Central" "Estadio Gran Parque
## $ City : chr [1:964] "Montevideo" "Montevideo" "Montevideo" "Montevideo" ...
## $ Country : chr [1:964] "Uruguay" "Uruguay" "Uruguay" "Uruguay" ...
## $ TournamentName : chr [1:964] "1930 FIFA World Cup" "1930 FIFA World Cup" "1930 FIFA World Cup" "1
## - attr(*, "spec")=
## .. cols(
## .. MatchID = col_character(),
## .. TournamentID = col_character(),
## .. MatchName = col_character(),
## .. Stage = col_character(),
## .. MatchDate = col_character(),
## .. MatchTime = col_time(format = ""),
## .. StadiumID = col_character(),
## .. HomeTeamID = col_character(),
## .. AwayTeamID = col_character(),
## .. HomeTeamScore = col_double(),
## .. AwayTeamScore = col_double(),
## .. ExtraTime = col_double(),
## .. PenaltyShootout = col_double(),
## .. HomePenalty = col_character(),
## .. AwayPenalty = col_character(),
## .. Result = col_character()
## .. )
## - attr(*, "problems")=<externalptr>
```

Display a summary of the main variables.

```
summary(
  df[c(
    "Result",
    "Stage",
    "Country",
    "ExtraTime",
    "PenaltyShootout",
    "HomePenalty",
    "AwayPenalty"
  )]
)
```

##	Result	Stage	Country	ExtraTime
##	Length:964	Length:964	Length:964	Min. :0.00000
##	Class :character	Class :character	Class :character	1st Qu.:0.00000
##	Mode :character	Mode :character	Mode :character	Median :0.00000
##				Mean :0.07573
##				3rd Qu.:0.00000
##				Max. :1.00000
##	PenaltyShootout	HomePenalty	AwayPenalty	
##	Min. :0.00000	Length:964	Length:964	
##	1st Qu.:0.00000	Class :character	Class :character	
##	Median :0.00000	Mode :character	Mode :character	
##	Mean :0.03631			
##	3rd Qu.:0.00000			

```
## Max. :1.00000
```

```
# Convert categorical variables to factors.
df$Result <- as.factor(df$Result)
df$Stage <- as.factor(df$Stage)
df$Country <- as.factor(df$Country)
df$ExtraTime <- as.factor(df$ExtraTime)

# Convert penalty variables to numeric.
df$HomePenalty <- as.numeric(df$HomePenalty)
df$AwayPenalty <- as.numeric(df$AwayPenalty)

# Replace missing penalty values with zero.
df$HomePenalty[is.na(df$HomePenalty)] <- 0
df$AwayPenalty[is.na(df$AwayPenalty)] <- 0
```

Code Testing

```
# The number of rows should be unchanged after joining.
nrow(matches)
```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

```
# Check the classes after conversion.
class(df$Result)
```

```
## [1] "factor"
```

```
class(df$Stage)
```

```
## [1] "factor"
```

```
class(df$Country)
```

```
## [1] "factor"
```

```
class(df$ExtraTime)
```

```
## [1] "factor"
```

```
class(df$HomePenalty)
```

```
## [1] "numeric"
```

```
class(df$AwayPenalty)
```

```
## [1] "numeric"
```

```
# Check that no missing penalty values remain.
```

```
sum(is.na(df$HomePenalty))
```

```
## [1] 0
```

```
sum(is.na(df$AwayPenalty))
```

```
## [1] 0
```

The number of rows in `matches` and `df` should be the same. This is important because joining the other datasets should add more columns, but it should not remove or add match records.

The four category variables should return `"factor"`. The penalty variables should return `"numeric"`. The two missing-value checks should return zero. These checks help me confirm that the data was changed into the correct format before doing the analysis.

Results and Interpretation

```
# Keep only matches involving a penalty shootout.
```

```
penalty_matches <- df %>%  
  filter(PenaltyShootout == 1)
```

```
# Create a frequency table for HomePenalty.
```

```
penalty_table <- table(penalty_matches$HomePenalty)  
penalty_table
```

```
##
```

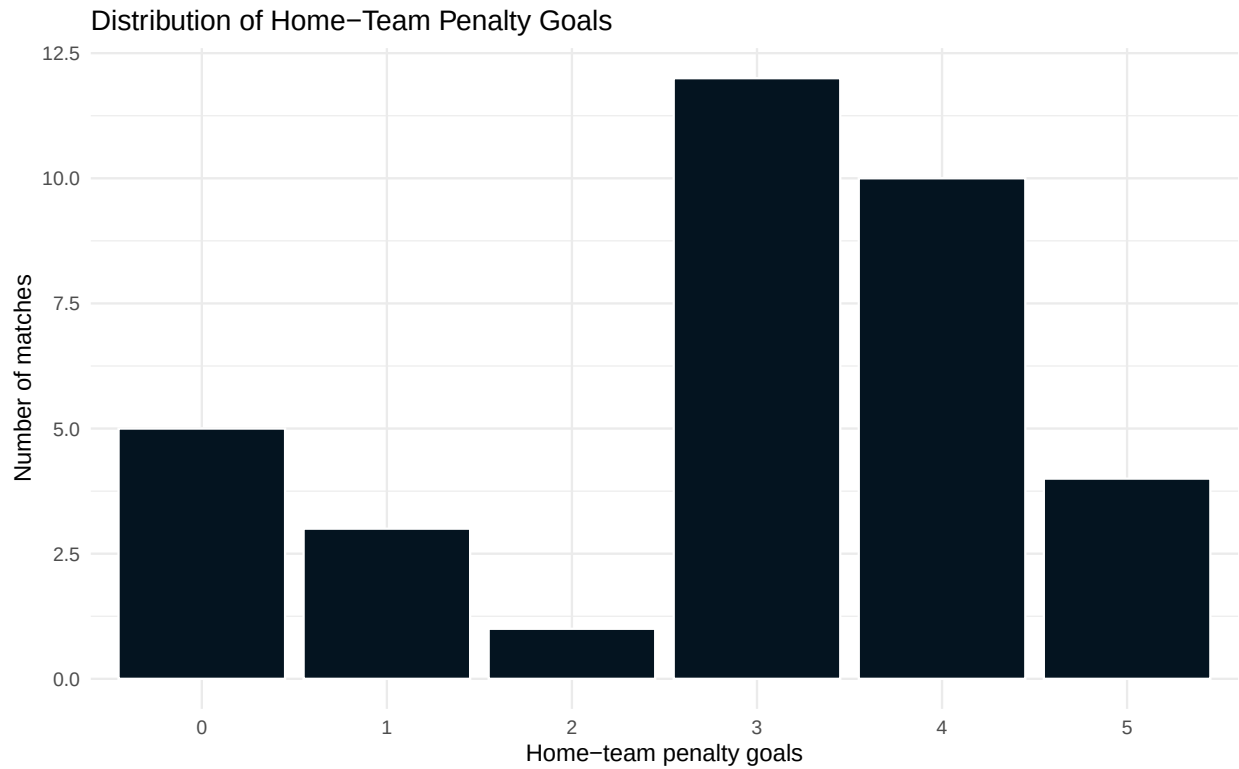
```
## 0  1  2  3  4  5
```

```
## 5  3  1 12 10  4
```

```
# Find the most common HomePenalty value.
```

```
most_common_penalty <- names(penalty_table)[which.max(penalty_table)]  
most_common_count <- max(penalty_table)
```

```
ggplot(penalty_matches, aes(x = factor(HomePenalty))) +  
  geom_bar(  
    fill = darkest_colour,  
    colour = "white"  
  ) +  
  labs(  
    title = "Distribution of Home-Team Penalty Goals",  
    x = "Home-team penalty goals",  
    y = "Number of matches"  
  ) +  
  theme_minimal()
```



There were 35 matches that included a penalty shootout. The most common value of `HomePenalty` was 3. This value appeared in 12 matches.

From the chart, some penalty scores appeared more often than the other scores. The taller bars mean that the score happened in more matches. The shorter bars mean that the score was less common. The distribution was not equal because the bars did not have the same height.

Question 2: Distribution of Home Team Scores

Rationale

I used histograms because `HomeTeamScore` is a numerical variable. Football scores are whole numbers, so I used a bin width of one. This means that each bar represents one score, such as 0, 1, 2 or 3 goals.

I compared the home-team scores by `Stage`, `TournamentName` and `AwayScoreGroup`. I separated these comparisons because putting all the information in one chart would make the chart too crowded and difficult to read.

I also used `aggregate()` to calculate the mean and median for each group. The histograms show the shape of the score distribution, while the mean and median give simple numbers that can be compared between groups.

Distribution by Competition Stage

```
# Calculate the mean and median HomeTeamScore for each stage.
stage_mean <- aggregate(
  HomeTeamScore ~ Stage,
  mean,
  data = df
)

stage_median <- aggregate(
  HomeTeamScore ~ Stage,
  median,
  data = df
)

stage_mean
```

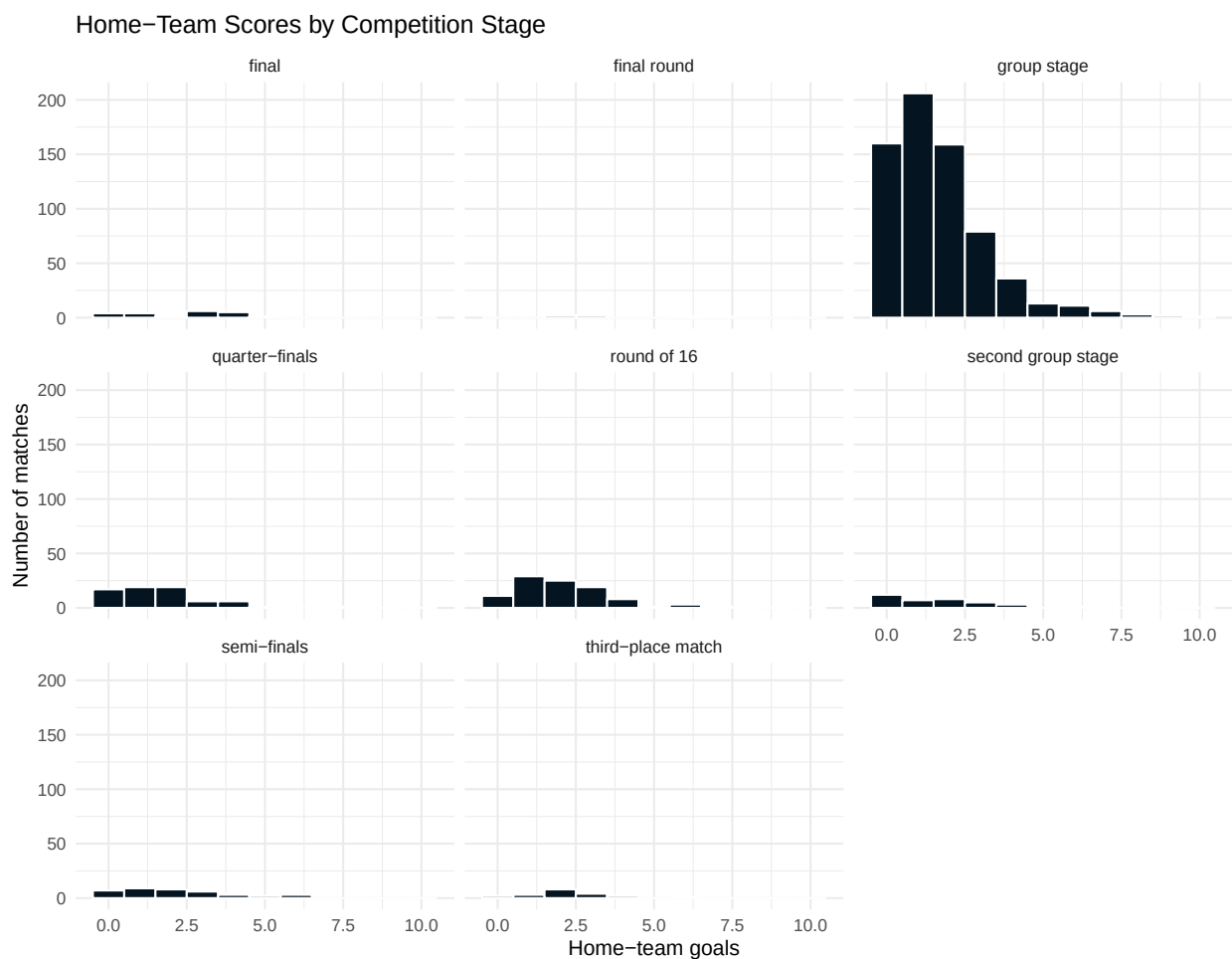
```
##           Stage HomeTeamScore
## 1           final      2.333333
## 2    final round      3.833333
## 3    group stage      1.671598
## 4 quarter-finals      1.700000
## 5    round of 16      2.041237
## 6 second group stage      1.555556
## 7    semi-finals      2.184211
## 8 third-place match      2.250000
```

```
stage_median
```

```
##           Stage HomeTeamScore
## 1           final              3
## 2    final round              3
## 3    group stage              1
## 4 quarter-finals              1
## 5    round of 16              2
## 6 second group stage          1
## 7    semi-finals              2
## 8 third-place match          2
```



```
ggplot(df, aes(x = HomeTeamScore)) +
  geom_histogram(
    binwidth = 1,
    fill = darkest_colour,
    colour = "white"
  ) +
  facet_wrap(~ Stage) +
  labs(
    title = "Home-Team Scores by Competition Stage",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```



Distribution by Tournament Name

```
# Calculate the mean and median HomeTeamScore for each tournament.
tournament_mean <- aggregate(
  HomeTeamScore ~ TournamentName,
  mean,
```

```

data = df
)

tournament_median <- aggregate(
  HomeTeamScore ~ TournamentName,
  median,
  data = df
)

tournament_mean

```

```

##      TournamentName HomeTeamScore
## 1  1930 FIFA World Cup      3.277778
## 2  1934 FIFA World Cup      2.823529
## 3  1938 FIFA World Cup      3.388889
## 4  1950 FIFA World Cup      3.136364
## 5  1954 FIFA World Cup      4.192308
## 6  1958 FIFA World Cup      2.514286
## 7  1962 FIFA World Cup      2.156250
## 8  1966 FIFA World Cup      2.156250
## 9  1970 FIFA World Cup      2.218750
## 10 1974 FIFA World Cup      1.342105
## 11 1978 FIFA World Cup      1.894737
## 12 1982 FIFA World Cup      1.846154
## 13 1986 FIFA World Cup      1.384615
## 14 1990 FIFA World Cup      1.230769
## 15 1994 FIFA World Cup      1.596154
## 16 1998 FIFA World Cup      1.531250
## 17 2002 FIFA World Cup      1.390625
## 18 2006 FIFA World Cup      1.343750
## 19 2010 FIFA World Cup      1.187500
## 20 2014 FIFA World Cup      1.265625
## 21 2018 FIFA World Cup      1.421875
## 22 2022 FIFA World Cup      1.578125

```

```

tournament_median

```

```

##      TournamentName HomeTeamScore
## 1  1930 FIFA World Cup          3.0
## 2  1934 FIFA World Cup          3.0
## 3  1938 FIFA World Cup          3.0
## 4  1950 FIFA World Cup          2.5
## 5  1954 FIFA World Cup          4.0
## 6  1958 FIFA World Cup          2.0
## 7  1962 FIFA World Cup          2.0
## 8  1966 FIFA World Cup          2.0
## 9  1970 FIFA World Cup          2.0
## 10 1974 FIFA World Cup          1.0
## 11 1978 FIFA World Cup          2.0
## 12 1982 FIFA World Cup          1.0
## 13 1986 FIFA World Cup          1.0
## 14 1990 FIFA World Cup          1.0
## 15 1994 FIFA World Cup          1.0

```

```
## 16 1998 FIFA World Cup      1.0
## 17 2002 FIFA World Cup      1.0
## 18 2006 FIFA World Cup      1.0
## 19 2010 FIFA World Cup      1.0
## 20 2014 FIFA World Cup      1.0
## 21 2018 FIFA World Cup      1.0
## 22 2022 FIFA World Cup      1.0
```

```
ggplot(df, aes(x = HomeTeamScore)) +
  geom_histogram(
    binwidth = 1,
    fill = darkest_colour,
    colour = "white"
  ) +
  facet_wrap(~ TournamentName) +
  labs(
    title = "Home-Team Scores by Tournament",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```

Home-Team Scores by Tournament



Distribution by Away-Team Score Group

```
# Create an empty variable for the away-team score groups.
df$AwayScoreGroup <- NA
```

```

# Assign each match to one of the three required groups.
df$AwayScoreGroup[df$AwayTeamScore <= 1] <- "0-1 goals"

df$AwayScoreGroup[
  df$AwayTeamScore >= 2 &
  df$AwayTeamScore <= 3
] <- "2-3 goals"

df$AwayScoreGroup[df$AwayTeamScore >= 4] <- "4 or more goals"

# Convert the new variable to a factor.
df$AwayScoreGroup <- as.factor(df$AwayScoreGroup)

# Check the number of matches in each group.
table(df$AwayScoreGroup)

```

```

##
##      0-1 goals      2-3 goals 4 or more goals
##           701           237           26

```

```

# Count the number of matches in each group.
away_group_counts <- table(df$AwayScoreGroup)

# Put the groups in order from the smallest count to the largest count.
away_group_order <- names(sort(away_group_counts))

# Give a light colour to the smallest group and a dark colour to the largest group.
away_group_colours <- setNames(
  c(
    lightest_colour,
    middle_colour,
    darkest_colour
  ),
  away_group_order
)

# Calculate the mean and median HomeTeamScore for each group.
away_group_mean <- aggregate(
  HomeTeamScore ~ AwayScoreGroup,
  mean,
  data = df
)

away_group_median <- aggregate(
  HomeTeamScore ~ AwayScoreGroup,
  median,
  data = df
)

away_group_mean

```

```

##      AwayScoreGroup HomeTeamScore

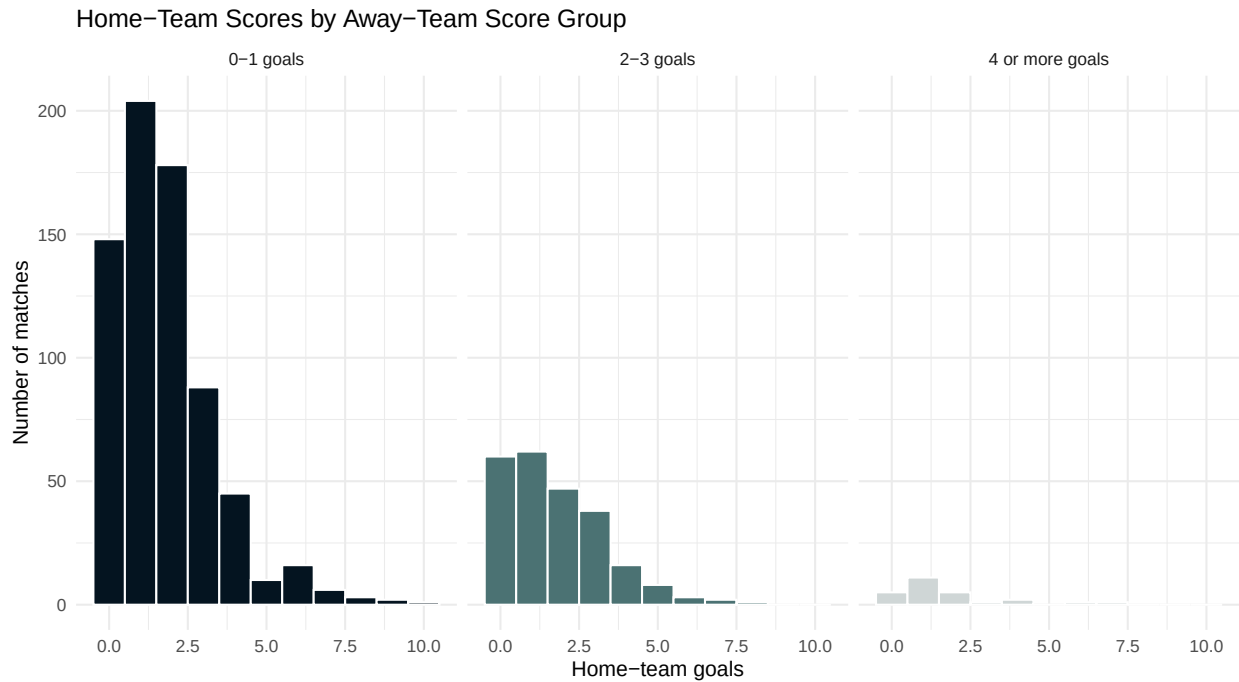
```

```
## 1      0-1 goals      1.774608
## 2      2-3 goals      1.746835
## 3 4 or more goals      1.730769
```

```
away_group_median
```

```
##      AwayScoreGroup HomeTeamScore
## 1      0-1 goals      1
## 2      2-3 goals      1
## 3 4 or more goals      1
```

```
ggplot(
  df,
  aes(
    x = HomeTeamScore,
    fill = AwayScoreGroup
  )
) +
  geom_histogram(
    binwidth = 1,
    colour = "white"
  ) +
  facet_wrap(~ AwayScoreGroup) +
  scale_fill_manual(
    values = away_group_colours
  ) +
  guides(fill = "none") +
  labs(
    title = "Home-Team Scores by Away-Team Score Group",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```



Code Testing

```
# Check the score variable types.
class(df$HomeTeamScore)
```

```
## [1] "numeric"
```

```
class(df$AwayTeamScore)
```

```
## [1] "numeric"
```

```
# Check for missing score values.
sum(is.na(df$HomeTeamScore))
```

```
## [1] 0
```

```
sum(is.na(df$AwayTeamScore))
```

```
## [1] 0
```

```
# Check the new group variable.
class(df$AwayScoreGroup)
```

```
## [1] "factor"
```

```
table(df$AwayScoreGroup)
```

```
##  
##      0-1 goals      2-3 goals 4 or more goals  
##      701          237          26
```

```
sum(is.na(df$AwayScoreGroup))
```

```
## [1] 0
```

```
# Check that all matches are included in the three groups.  
sum(table(df$AwayScoreGroup))
```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

HomeTeamScore and AwayTeamScore should be numeric variables. R may show "integer" because football scores are whole numbers, and this is still suitable for this analysis.

The missing-value results should be zero. AwayScoreGroup should be a factor and it should not contain missing values. The total number of matches in the three groups should also be equal to the number of rows in df. This check confirms that every match was placed into one group.

Results and Interpretation

The stage with the highest average home-team score was **final round**, with an average of 3.83 goals. The stage with the lowest average was **second group stage**, with an average of 1.56 goals.

The histograms show that most home-team scores were around the lower score values. Some stages had more matches than the other stages, so their bars were also taller. The chart is useful for comparing the general score pattern, but the mean is also needed because the number of matches was different between stages.

The tournament with the highest average home-team score was **1954 FIFA World Cup**, with 4.19 goals. The tournament with the lowest average was **2010 FIFA World Cup**, with 1.19 goals.

The score distribution was not exactly the same for every tournament. Some tournaments had more high-scoring matches, while some tournaments had more scores around zero, one or two goals. However, the number of matches was also different between tournaments, so the charts should be compared together with the mean and median.

For the away-team score groups, **0-1 goals** had the highest average home-team score, at 1.77 goals. This shows that the home-team score was different between the three away-score groups. However, this is only a relationship found in the dataset. It does not mean that the away-team score directly caused the home-team score.

Question 3: Penalty Shootout Participation

Rationale

I first selected only the matches where `PenaltyShootout` was equal to 1. I did this because Question 3 only asks about matches that included a penalty shootout.

After filtering the matches, I grouped the data by the team name and team code. I counted the number of matches for each team by using `summarise()` and `n()`.

I did the home-team and away-team analysis separately because a team may be recorded as the home team in one match and the away team in another match. I then sorted the results from the highest number to the lowest number. Finally, I used `head()` to select the first five teams.

Filter Penalty Shootout Matches

```
# Keep only matches that involved a penalty shootout.
penalty_matches <- df %>%
  filter(PenaltyShootout == 1)

# Check the number of penalty-shootout matches.
nrow(penalty_matches)
```

```
## [1] 35
```

```
# Check that the filtered data contains only value 1.
table(penalty_matches$PenaltyShootout)
```

```
##
##  1
## 35
```

Top Five Home Teams

```
# Group the penalty matches by home-team name and code.
# Count the number of matches for each home team.
home_penalty_count <- penalty_matches %>%
  group_by(HomeTeamName, HomeTeamCode) %>%
  summarise(NumberOfMatches = n())

# Sort the teams from the highest count to the lowest count.
home_penalty_count <- home_penalty_count %>%
  arrange(desc(NumberOfMatches))

# Select the first five teams.
top5_home <- head(home_penalty_count, 5)

# Change the column names to match the assignment requirement.
names(top5_home) <- c(
```

```

  "TeamName",
  "TeamCode",
  "NumberOfMatches"
)

# Display the top five home teams as a table.
top5_home

```

```

## # A tibble: 5 x 3
## # Groups:   TeamName [5]
##   TeamName    TeamCode NumberOfMatches
##   <chr>      <chr>          <int>
## 1 Argentina    ARG              4
## 2 Brazil       BRA              4
## 3 Spain        ESP              4
## 4 Netherlands  NLD              3
## 5 West Germany DEU              3

```

Top Five Away Teams

```

# Group the penalty matches by away-team name and code.
# Count the number of matches for each away team.
away_penalty_count <- penalty_matches %>%
  group_by(AwayTeamName, AwayTeamCode) %>%
  summarise(NumberOfMatches = n())

# Sort the teams from the highest count to the lowest count.
away_penalty_count <- away_penalty_count %>%
  arrange(desc(NumberOfMatches))

# Select the first five teams.
top5_away <- head(away_penalty_count, 5)

# Change the column names to match the assignment requirement.
names(top5_away) <- c(
  "TeamName",
  "TeamCode",
  "NumberOfMatches"
)

# Display the top five away teams as a table.
top5_away

```

```

## # A tibble: 5 x 3
## # Groups:   TeamName [5]
##   TeamName    TeamCode NumberOfMatches
##   <chr>      <chr>          <int>
## 1 France     FRA              5
## 2 Argentina  ARG              3
## 3 England    ENG              3
## 4 Croatia    HRV              2
## 5 Italy       ITA              2

```

Code Testing

```
# Check that all filtered matches involved a penalty shootout.  
table(penalty_matches$PenaltyShootout)
```

```
##  
## 1  
## 35
```

```
# Both top-five tables should contain five rows.  
nrow(top5_home)
```

```
## [1] 5
```

```
nrow(top5_away)
```

```
## [1] 5
```

```
# Check that both tables have the required column names.  
names(top5_home)
```

```
## [1] "TeamName"      "TeamCode"      "NumberOfMatches"
```

```
names(top5_away)
```

```
## [1] "TeamName"      "TeamCode"      "NumberOfMatches"
```

```
# Check for missing values in team names and codes.  
sum(is.na(top5_home$TeamName))
```

```
## [1] 0
```

```
sum(is.na(top5_home$TeamCode))
```

```
## [1] 0
```

```
sum(is.na(top5_away$TeamName))
```

```
## [1] 0
```

```
sum(is.na(top5_away$TeamCode))
```

```
## [1] 0
```

```
# Check that the match counts are sorted from high to low.
top5_home$NumberOfMatches
```

```
## [1] 4 4 4 3 3
```

```
top5_away$NumberOfMatches
```

```
## [1] 5 3 3 2 2
```

The `PenaltyShootout` table should only show the value 1. This confirms that the filtered data only contains penalty-shootout matches.

Both final tables should contain five rows. They should also contain the columns `TeamName`, `TeamCode` and `NumberOfMatches`. The missing-value checks should return zero. The match counts should be shown from the highest value to the lowest value.

Results and Interpretation

The first team in the home-team table was **Argentina**. This team appeared in 4 penalty-shootout matches as the home team.

The first team in the away-team table was **France**. This team appeared in 5 penalty-shootout matches as the away team.

There were 1 teams that appeared in both top-five tables. These teams were **Argentina**. The two rankings were **different** when I compared the team names and their order.

The home and away rankings can be different because the home-team and away-team labels can change from one match to another match. In this dataset, being called the home team also does not always mean that the team played in its own country.

Question 4: Factors Influencing Match Outcomes

Selected Factors and Rationale

I selected `Stage` and `ExtraTime` as the two factors.

I selected `Stage` because matches may be played differently at different parts of the tournament. For example, a group-stage match may not have the same situation as a semi-final or final match. Teams may use a different strategy because the importance of the match is different.

I selected `ExtraTime` because matches that continue after regular time may have a different result pattern. Extra time may happen when the teams are close in performance, so I wanted to compare these matches with the other matches.

I used frequency tables first because they show the number of home wins, away wins and draws. I also used row percentages because the number of matches was not the same in every group. The proportional bar charts make the three result categories easier to compare.

Overall Match Results

```
# Count the number of matches in each Result category.
overall_result_table <- table(df$Result)
```

```
overall_result_table
```

```
##
## away team win      draw home team win
##           240           179           545
```

```
# Put the Result categories in order
# from the smallest count to the largest count.
result_order <- names(sort(overall_result_table))
```

```
# Give a light colour to the least common result
# and a dark colour to the most common result.
```

```
result_colours <- setNames(
  c(
    lightest_colour,
    middle_colour,
    darkest_colour
  ),
  result_order
)
```

```
# Find the most common match result.
```

```
most_common_result <- names(overall_result_table)[
  which.max(overall_result_table)
]
```

```
most_common_result_count <- max(overall_result_table)
```

Factor 1: Competition Stage

```
# Create a table of Stage and Result.
```

```
stage_result_count <- table(  
  df$Stage,  
  df$Result  
)
```

```
stage_result_count
```

```
##  
##               away team win draw home team win  
##   final               4     0             17  
##   final round          0     1             5  
##   group stage        164   166           346  
##   quarter-finals      24     2            44  
##   round of 16         21     2            74  
##   second group stage   11     8            17  
##   semi-finals         10     0            28  
##   third-place match    6     0            14
```

```
# Convert the counts into row percentages.
```

```
# margin = 1 calculates percentages within each Stage.
```

```
stage_result_percent <- prop.table(  
  stage_result_count,  
  margin = 1  
) * 100
```

```
# Round the percentages to one decimal place.
```

```
stage_result_percent <- round(  
  stage_result_percent,  
  1  
)
```

```
stage_result_percent
```

```
##  
##               away team win draw home team win  
##   final               19.0  0.0             81.0  
##   final round          0.0 16.7             83.3  
##   group stage        24.3 24.6             51.2  
##   quarter-finals      34.3  2.9             62.9  
##   round of 16         21.6  2.1             76.3  
##   second group stage   30.6 22.2             47.2  
##   semi-finals         26.3  0.0             73.7  
##   third-place match    30.0  0.0             70.0
```

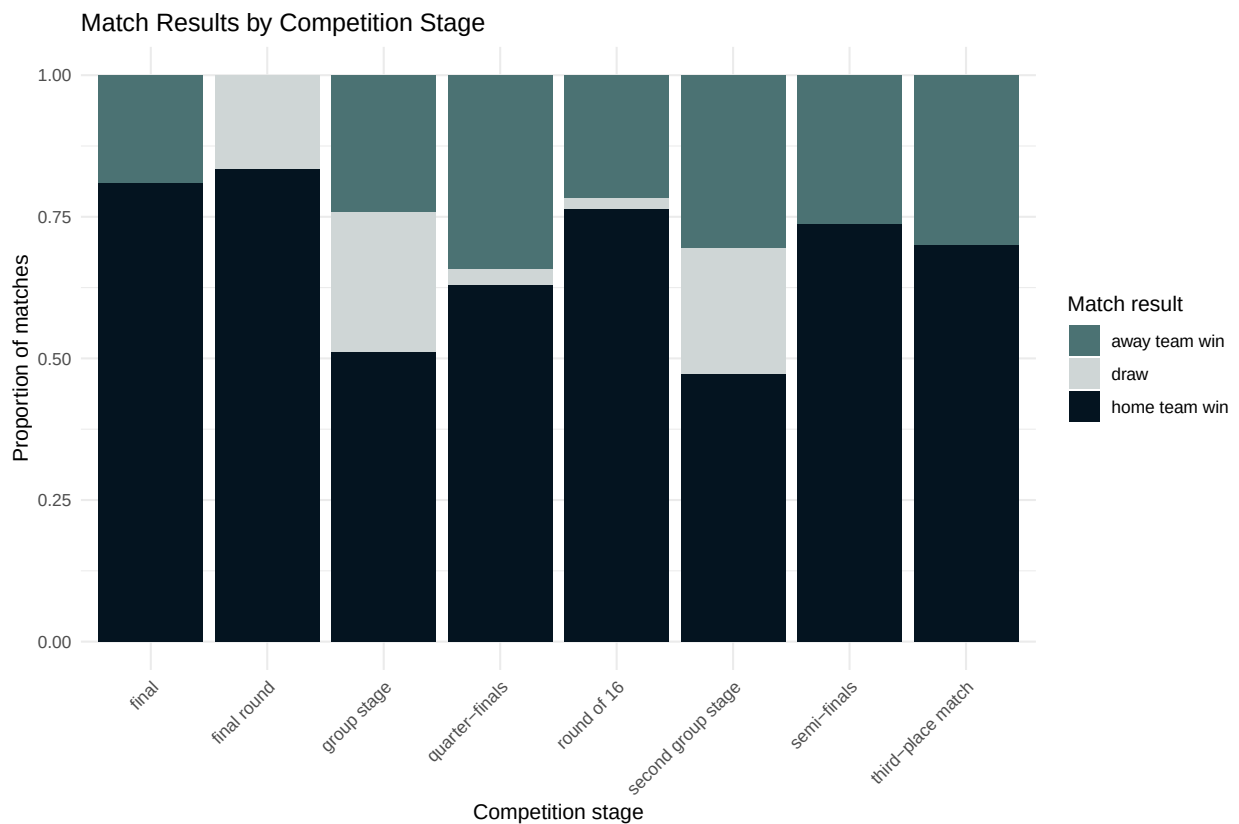
```
# Create a proportional bar chart of Result for each Stage.
```

```
ggplot(  
  df,  
  aes(  
    x = Stage,
```

```

    fill = Result
  )
) +
  geom_bar(position = "fill") +
  scale_fill_manual(
    values = result_colours
  ) +
  labs(
    title = "Match Results by Competition Stage",
    x = "Competition stage",
    y = "Proportion of matches",
    fill = "Match result"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 45,
      hjust = 1
    )
  )
)

```



Factor 2: Extra Time

```
# Create a table of ExtraTime and Result.
```

```
extra_time_result_count <- table(
  df$ExtraTime,
  df$Result
)
```

```
extra_time_result_count
```

```
##
```

```
##      away team win draw home team win
```

```
##    0          215  173          503
```

```
##    1           25   6           42
```

```
# Convert the counts into row percentages.
```

```
# margin = 1 calculates percentages within each ExtraTime group.
```

```
extra_time_result_percent <- prop.table(
  extra_time_result_count,
  margin = 1
) * 100
```

```
# Round the percentages to one decimal place.
```

```
extra_time_result_percent <- round(
  extra_time_result_percent,
  1
)
```

```
extra_time_result_percent
```

```
##
```

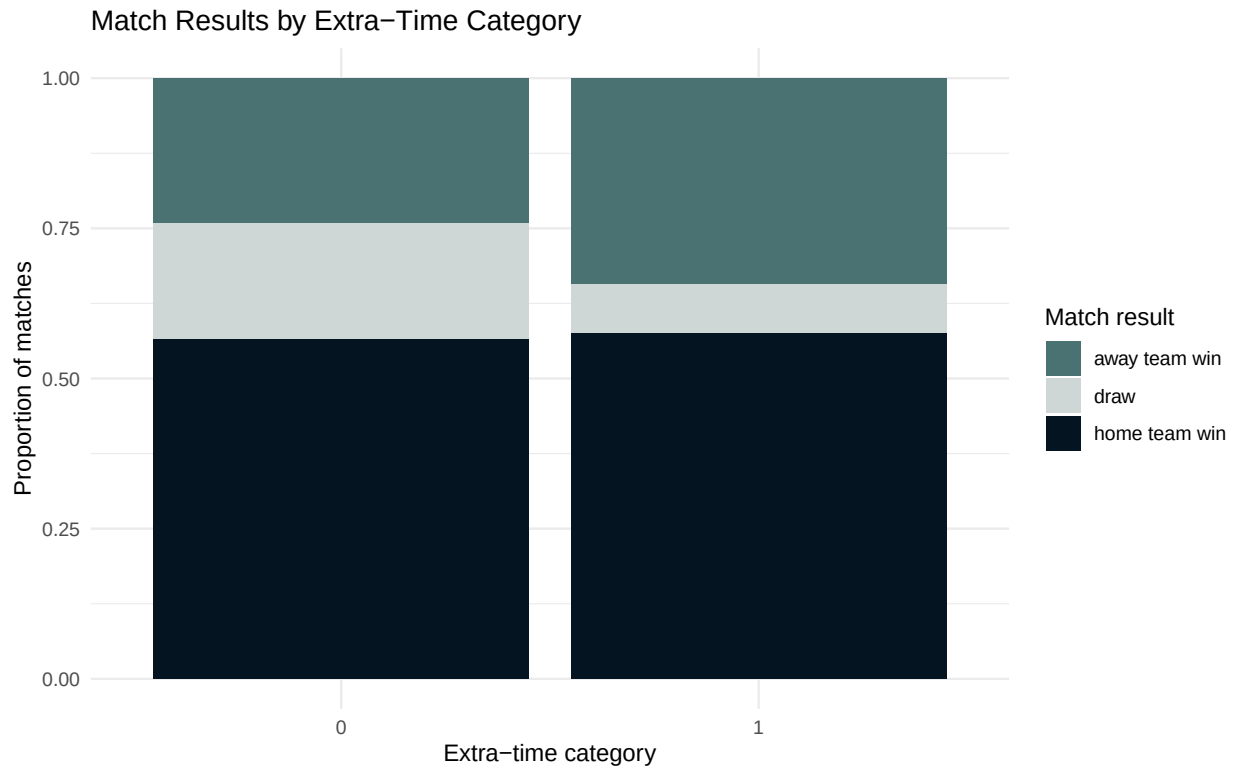
```
##      away team win draw home team win
```

```
##    0          24.1 19.4          56.5
```

```
##    1          34.2  8.2          57.5
```

```
# Create a proportional bar chart of Result by ExtraTime.
```

```
ggplot(
  df,
  aes(
    x = ExtraTime,
    fill = Result
  )
) +
  geom_bar(position = "fill") +
  scale_fill_manual(
    values = result_colours
  ) +
  labs(
    title = "Match Results by Extra-Time Category",
    x = "Extra-time category",
    y = "Proportion of matches",
    fill = "Match result"
  ) +
  theme_minimal()
```

Code Testing

```
# Check that the three variables are factors.
class(df$Result)
```

```
## [1] "factor"
```

```
class(df$Stage)
```

```
## [1] "factor"
```

```
class(df$ExtraTime)
```

```
## [1] "factor"
```

```
# Display the categories in each variable.
levels(df$Result)
```

```
## [1] "away team win" "draw"          "home team win"
```

```
levels(df$Stage)
```

```
## [1] "final"           "final round"     "group stage"
## [4] "quarter-finals" "round of 16"     "second group stage"
## [7] "semi-finals"     "third-place match"
```

```
levels(df$ExtraTime)
```

```
## [1] "0" "1"
```

```
# Check for missing values.
```

```
sum(is.na(df$Result))
```

```
## [1] 0
```

```
sum(is.na(df$Stage))
```

```
## [1] 0
```

```
sum(is.na(df$ExtraTime))
```

```
## [1] 0
```

```
# The total number in each table should equal the number of rows in df.
```

```
sum(stage_result_count)
```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

```
sum(extra_time_result_count)
```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

```
# Each row of the percentage tables should total approximately 100%.
```

```
rowSums(stage_result_percent)
```

```
##          final          final round      group stage    quarter-finals
##          100.0          100.0          100.1          100.1
## round of 16 second group stage    semi-finals third-place match
##          100.0          100.0          100.0          100.0
```

```
rowSums(extra_time_result_percent)
```

```
##      0      1
## 100.0  99.9
```

Result, Stage and ExtraTime should return "factor". Their levels show the categories stored inside each variable.

The missing-value checks should return zero. The total value from each count table should be equal to the number of rows in df. The rows in the percentage tables should add to around 100%. A result such as 99.9% or 100.1% can happen because the percentages were rounded.

Results and Interpretation

For all matches, the most common result was **home team win**. This result happened in 545 matches.

For the competition stages, **final round** had the highest percentage of home-team wins, at 83.3%. The percentage table shows that the result pattern was not exactly the same for every stage. Some stages had a higher percentage of home-team wins, while some stages had more away-team wins or draws.

For extra time, the category **1** had the highest percentage of home-team wins, at 57.5%. The chart also shows that the result distribution was different between the extra-time categories.

These results show that **Stage** and **ExtraTime** have a relationship with match results in this dataset. However, the analysis does not prove that these two factors directly caused a home-team win, away-team win or draw. Other factors may also affect the match result.

Question 5: Git and GitHub

GitHub Repository

I created and managed the submission repository by using Git commands in Ubuntu Terminal.

The repository is available at:

<https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission>

The analysis for Questions 1 to 4 was already prepared before this repository was created. I then organised the completed files into a new project folder and used Terminal commands to create the repository, make separate commits and push the project to GitHub.

Project Folder Structure

The project was organised into separate folders:

- **data/** contains the four FIFA World Cup datasets.
- **scripts/** contains the R scripts for Questions 1 to 4.
- **report/** contains the R Markdown file and the final PDF report.
- **figures/** contains the Terminal screenshots used as evidence.
- **COMP1013-FIFA-WorldCup-Analysis.Rproj** is the RStudio project file.

The folders keep the datasets, scripts, report and figures separate. This makes the project easier to understand and check.

Terminal Git Workflow Evidence

Preparing the New Project

```
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ pwd
/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ ls -la
ls data
ls scripts
ls report
ls figures
total 4
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57 .
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:56 ..
-rwxrwxrwx 1 yvette yvette 719 Jul 20 02:57 .gitignore
-rwxrwxrwx 1 yvette yvette 218 Jul 20 02:57 COMP1013-FIFA-WorldCup-Analysis.Rproj
-rwxrwxrwx 1 yvette yvette 139 Jul 20 02:57 README.md
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57 data
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57 figures
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57 report
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57 scripts
Matches.csv Stadiums.csv Teams.csv Tournaments.csv
01_data_preparation.R 03_penalty_shootouts.R
02_score_distribution.R 04_match_outcomes.R
Report.Rmd Report.pdf
commit_history.png
```

This screenshot shows the new submission folder and its project structure. The datasets, scripts, report and figures were placed into separate folders before Git was started.

Initialising Git

```
Initialized empty Git repository in /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission/.git/
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git branch -M main
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        COMP1013-FIFA-WorldCup-Analysis.Rproj
        README.md
        data/
        figures/
        report/
        scripts/

nothing added to commit but untracked files present (use "git add" to track
)
```

I used `git init` to create a new local Git repository. I also changed the main branch name to `main`. The `git status` result showed that the repository had no commits yet and the project files were still untracked.

First Commit: Project Configuration

```
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git add .gitignore
git add README.md
git add COMP1013-FIFA-WorldCup-Analysis.Rproj
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   .gitignore
    new file:   COMP1013-FIFA-WorldCup-Analysis.Rproj
    new file:   README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    data/
    figures/
    report/
    scripts/

yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git diff --cached --stat
.gitignore | 52 ++++++
COMP1013-FIFA-WorldCup-Analysis.Rproj | 13 +++++
README.md | 2 +
3 files changed, 67 insertions(+)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git commit -m "Set up project"
[main (root-commit) a25e784] Set up project
3 files changed, 67 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 COMP1013-FIFA-WorldCup-Analysis.Rproj
 create mode 100644 README.md
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git log --oneline --decorate
a25e784 (HEAD -> main) Set up project
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git status
On branch main

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    data/
    figures/
    report/
```

I used `git add` to stage the project configuration files. I then used `git commit` to create the first commit with the message **Set up project**.

Adding the Datasets

```
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ ls -la data
total 128
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57 
drwxrwxrwx 1 yvette yvette 512 Jul 20 03:01 
-rwxrwxrwx 1 yvette yvette 107135 Jul 20 02:57 Matches.csv
-rwxrwxrwx 1 yvette yvette 8299 Jul 20 02:57 Stadiums.csv
-rwxrwxrwx 1 yvette yvette 1595 Jul 20 02:57 Teams.csv
-rwxrwxrwx 1 yvette yvette 696 Jul 20 02:57 Tournaments.csv
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git add data/*.csv
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   data/Matches.csv
        new file:   data/Stadiums.csv
        new file:   data/Teams.csv
        new file:   data/Tournaments.csv

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        figures/
        report/
        scripts/

yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git diff --cached --stat
data/Matches.csv      | 965 +++++
data/Stadiums.csv     | 194 +++++
data/Teams.csv        | 86 +++++
data/Tournaments.csv  | 24 ++
4 files changed, 1269 insertions(+)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git commit -m "Add FIFA World Cup datasets"
[main a976ebb] Add FIFA World Cup datasets
4 files changed, 1269 insertions(+)
 create mode 100644 data/Matches.csv
 create mode 100644 data/Stadiums.csv
 create mode 100644 data/Teams.csv
 create mode 100644 data/Tournaments.csv
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git log --oneline --decorate
a976ebb (HEAD -> main) Add FIFA World Cup datasets
a25e784 Set up project
```

I added the four CSV datasets separately. The commit message **Add FIFA World Cup datasets** explains the files included in this step.

Adding the R Scripts

```
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ ls -la scripts
total 16
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57
drwxrwxrwx 1 yvette yvette 512 Jul 20 03:01
-rwxrwxrwx 1 yvette yvette 1950 Jul 20 02:57 01_data_preparation.R
-rwxrwxrwx 1 yvette yvette 2361 Jul 20 02:57 02_score_distribution.R
-rwxrwxrwx 1 yvette yvette 2010 Jul 20 02:57 03_penalty_shootouts.R
-rwxrwxrwx 1 yvette yvette 3132 Jul 20 02:57 04_match_outcomes.R
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git add scripts/*.R
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   scripts/01_data_preparation.R
    new file:   scripts/02_score_distribution.R
    new file:   scripts/03_penalty_shootouts.R
    new file:   scripts/04_match_outcomes.R

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    figures/
    report/

yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git diff --cached --stat
scripts/01_data_preparation.R | 86 +++++
scripts/02_score_distribution.R | 113 +++++
scripts/03_penalty_shootouts.R | 80 +++++
scripts/04_match_outcomes.R | 136 +++++
4 files changed, 415 insertions(+)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git commit -m
"Add R scripts for Questions 1 to 4"
[main 10elc8a] Add R scripts for Questions 1 to 4
4 files changed, 415 insertions(+)
create mode 100644 scripts/01_data_preparation.R
create mode 100644 scripts/02_score_distribution.R
create mode 100644 scripts/03_penalty_shootouts.R
create mode 100644 scripts/04_match_outcomes.R
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git log --one
line --decorate
10elc8a (HEAD -> main) Add R scripts for Questions 1 to 4
a976ebb Add FIFA World Cup datasets
a25e784 Set up project
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    figures/
    report/

nothing added to commit but untracked files present (use "git add" to track)
```

I added the four R scripts used for Questions 1 to 4. These scripts were committed with the message **Add R scripts for Questions 1 to 4**.

Adding the R Markdown Report

```
yvette@dntvish: /mnt/c/User: × + ▾

nothing added to commit but untracked files present (use "git add" to track)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
ls -la report
total 368
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57
drwxrwxrwx 1 yvette yvette 512 Jul 20 03:01
-rwxrwxrwx 1 yvette yvette 29403 Jul 20 02:57 Report.Rmd
-rwxrwxrwx 1 yvette yvette 342000 Jul 20 02:57 Report.pdf
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git add report/Report.Rmd
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   report/Report.Rmd

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        figures/
        report/Report.pdf

yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git diff --cached --stat
report/Report.Rmd | 880 ++++++
1 file changed, 880 insertions(+)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git commit -m "Add R Markdown analysis report"
[main 86ae4f7] Add R Markdown analysis report
1 file changed, 880 insertions(+)
create mode 100644 report/Report.Rmd
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git log --oneline --decorate
86ae4f7 (HEAD -> main) Add R Markdown analysis report
10e1c8a Add R scripts for Questions 1 to 4
a976ebb Add FIFA World Cup datasets
a25e784 Set up project
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        figures/
        report/Report.pdf

nothing added to commit but untracked files present (use "git add" to track)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Ana
lysis-Submission$ |
```

I added `Report.Rmd` in a separate commit. This keeps the report source separate from the datasets and R scripts in the commit history.

Creating the GitHub Repository and Pushing

```
yvette@dntvish: /mnt/c/User: × + ▾
nothing added to commit but untracked files present (use "git add" to track
)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-World
Cup-Analysis-Submission$ gh auth status
github.com
✓ Logged in to github.com account yvette195 (/home/yvette/.config/gh/host
s.yml)
- Active account: true
- Git operations protocol: https
- Token: gho_*****
- Token scopes: 'gist', 'read:org', 'repo', 'workflow'
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-World
Cup-Analysis-Submission$ git remote -v
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-World
Cup-Analysis-Submission$ gh repo create COMP1013-FIFA-WorldCup-Analysis-Sub
mission \
  --public \
  --source=. \
  --remote=origin \
  --push
✓ Created repository yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission o
n github.com
https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission
✓ Added remote https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis
-Submission.git
Enumerating objects: 23, done.
Counting objects: 100% (23/23), done.
Delta compression using up to 8 threads
Compressing objects: 100% (22/22), done.
Writing objects: 100% (23/23), 34.31 KiB | 305.00 KiB/s, done.
Total 23 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission.
git
* [new branch]      HEAD -> main
branch 'main' set up to track 'origin/main'.
✓ Pushed commits to https://github.com/yvette195/COMP1013-FIFA-WorldCup-Ana
lysis-Submission.git
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-World
Cup-Analysis-Submission$ |
```

I used GitHub CLI in Ubuntu Terminal to create the new public repository. The local repository was connected to GitHub as `origin`, and the existing commits were pushed to the remote `main` branch.

Checking the GitHub Connection

```
yvette@dntvish: /mnt/c/Users/ yvette@ dntvish: /mnt/c/Users/ x + v
Total 23 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission.git
* [new branch] HEAD -> main
branch 'main' set up to track 'origin/main'.
✓ Pushed commits to https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission.git
yvette@dntvish: /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git remote -v

origin https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission.git (fetch)
origin https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission.git (push)
yvette@dntvish: /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
yvette@dntvish: /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git branch -vv
* main 86ae4f7 [origin/main] Add R Markdown analysis report
yvette@dntvish: /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git log --oneline --decorate -5
86ae4f7 (HEAD -> main, origin/main) Add R Markdown analysis report
10e1c8a Add R scripts for Questions 1 to 4
a976ebb Add FIFA World Cup datasets
a25e784 Set up project
yvette@dntvish: /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    figures/
    report/Report.pdf

nothing added to commit but untracked files present (use "git add" to track)
yvette@dntvish: /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ |
```

The `git remote -v` result shows the new GitHub repository URL. The branch information shows that the local `main` branch follows `origin/main`. The commit history also confirms that the local and remote repositories contain the same commits.

Adding and Pushing the Terminal Evidence

```
yvette@dntvish: /mnt/c/User: × + ∨

nothing added to commit but untracked files present (use "git add" to track)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git add figures/*.png
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   figures/commit_history.png
    new file:   figures/terminal_01_prepare_new_project.png
    new file:   figures/terminal_02_git_init.png
    new file:   figures/terminal_03_first_commit.png
    new file:   figures/terminal_04_add_datasets.png
    new file:   figures/terminal_05_add_scripts.png
    new file:   figures/terminal_06_add_rmarkdown_report.png
    new file:   figures/terminal_07_create_repo_and_push.png
    new file:   figures/terminal_08_verify_github_connection.png

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    report/Report.pdf

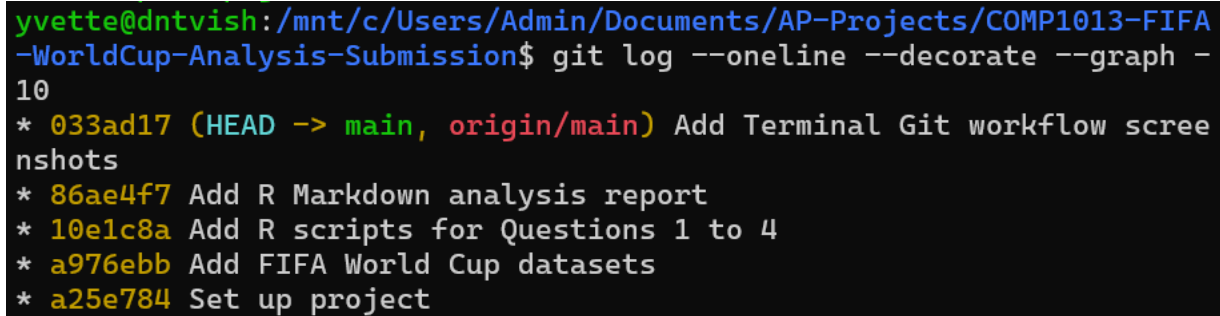
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git diff --cached --stat
figures/commit_history.png      | Bin 0 -> 90895 bytes
figures/terminal_01_prepare_new_project.png | Bin 0 -> 112459 bytes
figures/terminal_02_git_init.png | Bin 0 -> 60799 bytes
figures/terminal_03_first_commit.png | Bin 0 -> 127064 bytes
figures/terminal_04_add_datasets.png | Bin 0 -> 144668 bytes
figures/terminal_05_add_scripts.png | Bin 0 -> 157007 bytes
figures/terminal_06_add_rmarkdown_report.png | Bin 0 -> 161856 bytes
figures/terminal_07_create_repo_and_push.png | Bin 0 -> 162161 bytes
figures/terminal_08_verify_github_connection.png | Bin 0 -> 169469 bytes
9 files changed, 0 insertions(+), 0 deletions(-)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git commit -m "Add Terminal Git workflow screenshots"
[main 033ad17] Add Terminal Git workflow screenshots
9 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 figures/commit_history.png
create mode 100644 figures/terminal_01_prepare_new_project.png
create mode 100644 figures/terminal_02_git_init.png
create mode 100644 figures/terminal_03_first_commit.png
create mode 100644 figures/terminal_04_add_datasets.png
create mode 100644 figures/terminal_05_add_scripts.png
create mode 100644 figures/terminal_06_add_rmarkdown_report.png
create mode 100644 figures/terminal_07_create_repo_and_push.png
create mode 100644 figures/terminal_08_verify_github_connection.png
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git pull --rebase origin main
git push origin main
From https://github.com/yvettel195/COMP1013-FIFA-WorldCup-Analysis-Submission
 * branch      main       -> FETCH_HEAD
Current branch main is up to date.
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 8 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (12/12), 1.01 MiB | 7.25 MiB/s, done.
Total 12 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/yvettel195/COMP1013-FIFA-WorldCup-Analysis-Submission.git
 86ae4f7..033ad17  main -> main
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git status
git log --oneline --decorate -6
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    report/Report.pdf

nothing added to commit but untracked files present (use "git add" to track)
033ad17 (HEAD -> main, origin/main) Add Terminal Git workflow screenshots
86ae4f7 Add R Markdown analysis report
10elc8a Add R scripts for Questions 1 to 4
a976ebb Add FIFA World Cup datasets
a25e784 Set up project
```

I added the Terminal screenshots by using `git add`. I then committed them with the message **Add Terminal Git workflow screenshots** and pushed the new commit to GitHub.

Commit History



```
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git log --oneline --decorate --graph -10
* 033ad17 (HEAD -> main, origin/main) Add Terminal Git workflow screenshots
* 86ae4f7 Add R Markdown analysis report
* 10e1c8a Add R scripts for Questions 1 to 4
* a976ebb Add FIFA World Cup datasets
* a25e784 Set up project
```

The screenshot shows the commit history of the new submission repository. The commits were created separately for the project configuration, datasets, R scripts, R Markdown report and Terminal screenshots.

Explanation of the Commit History

The commit history shows that I added the project files through separate steps. I did not add all the files in only one commit.

1. **Set up project**

This commit added the `.gitignore`, `README` and RStudio project file.

2. **Add FIFA World Cup datasets**

This commit added the four CSV datasets used for the assignment.

3. **Add R scripts for Questions 1 to 4**

This commit added the four R scripts used for the data preparation and analysis.

4. **Add R Markdown analysis report**

This commit added the `Report.Rmd` source file.

5. **Add Terminal Git workflow screenshots**

This commit added the screenshots showing the Git and GitHub commands used in Ubuntu Terminal.

The commits separate the main parts of the project. The commit messages are simple and they show what was added at each step.

The screenshot was taken before the final report commit because the image needed to be inserted into the report before the final PDF could be created.